

Sea Journey Workflow — TNT Seal Management

End-to-end process from customer booking to billed completion, as implemented in `tnt_seal_management`. Five core doctypes drive the process; `Seal Journey` is the central record that every other doctype mirrors its status onto.

Step-by-Step Flow

1. Booking

- Customer (portal) or Account Manager creates a **Tagging Booking**. `booking_status = Draft` (or `Pending Account Manager Review` for portal bookings).
- A **Seal Journey** is created immediately and linked (`journey_status = Draft`).

2. Finance Approval

- Account Manager runs `submit_to_finance` → Booking `Pending Finance PCB Approval` ; Seal Journey mirrors the same status.
- Finance runs `approve_booking` → Booking `Finance PCB Approved` ; Seal Journey → `Finance PCB Approved` .
 - A **PCB Job Order** is created/reused, `job_order_status = Unassigned` .
 - (`reject_booking` → `Finance PCB Rejected` ; `reopen_booking` reverts to Draft.)

3. Team Lead & Technician Assignment

- PCB Job Order auto-assigns the sole **PCB Team Leader** on save → `job_order_status = Team Leader Assigned` ; Seal Journey → `Team Lead Assigned` .
- Team Leader runs `assign_to_field_technician` →
 - Creates/updates a **PCB Assignment** (`assignment_status = Assigned`).
 - Creates a **Journey Request** (`journey_request_status = Draft`).
 - Seal Journey → `Technician Assigned` .

4. Pre-Tagging & Submission to Control Room

- Field Technician fills the Journey Request: entry/departure card numbers, seals, pre-tagging checklist, tagging photos.
- Runs `submit_to_control_room` → Journey Request `Pending Control Room Approval` ; Seal Journey → `Pre-Tagging` ; seal custody moves **Warehouse** → **Technician**.
- `swap_seal` is available only at this stage if a seal needs replacing.

5. Control Room Approval

- Operations Control Room runs `approve_by_control_room` → Journey Request `Tagging` ; Seal Journey → `Tagging In Progress` .
- (`reject_by_control_room` → `Rejected` .)

6. Tagging → Journey Starts

- Field Technician runs `complete_tagging` (captures live GPS location) → Journey Request `Journey Ready` ; Seal Journey is finalized (vehicle, container, seals table, photos) and set directly to `In Transit` ; seal custody → **Customer**.
- There is no separate approval gate after tagging — the technician's own confirmation both closes tagging and starts the journey.

7. In Transit

- Live GPS/API telemetry (Uffizio integration) streams into the Seal Journey Transit tab via **Seal API Sync Log**.
- Anomalies raise **Seal Alert Log** entries (Critical/Warning/Info).

8. Arrival

- Operations Control Room runs `confirm_arrival` on the Seal Journey, choosing an unlock method:
 - **Physical unlock** → Seal Journey `Arrived` ; raises an **Untagging PCB Assignment**.
 - **Remote unlock** → Seal Journey skips straight to `Awaiting Seal Return` ; raises a **Seal Return PCB Assignment**.

9. Untagging (physical-unlock path only)

- Team Leader assigns a technician on the Untagging PCB Assignment → Journey Request `Untagging` ; Seal Journey → `Untagging In Progress` .
- Control Room runs `start_untagging` / `complete_untagging` on the Seal Journey → `Untagged` .
- Field Technician runs `confirm_untagging` on the Journey Request (evidence photos + confirmation) → Journey Request `Awaiting Seal Return` ; Seal Journey → `Awaiting Seal Return` .

10. Seal Return

- Field Technician runs `confirm_seal_return` on the Journey Request (seal condition, retrieval card number, return warehouse, evidence photos) → Journey Request `Seal Returned` ; Seal Journey → `Completed` ; Seal Device → `Returned` , custody → **Warehouse**.

11. Billing

- Billing is recomputed on every Seal Journey save via **Seal Billing Rate** (through **Customer Billing Assignment**): `billing_status = Not Billed` → `Pending Billing` (or `No Per-Journey Charge`).
- Finance generates an ERPNext Sales Order for completed, unbilled journeys (`generate_sales_order`) → `billing_status = Processing Payment` .
- On Sales Invoice payment (`sync_billing_from_sales_invoice` hook) → `billing_status = Billed` — the journey is fully closed out.

Status Reference

Seal Journey (`journey_status`) — the authoritative end-to-end status:

Draft
→ Pending Finance PCB Approval
→ Finance PCB Approved / Finance PCB Rejected
→ Team Lead Assigned
→ Technician Assigned
→ Pre-Tagging
→ Tagging In Progress
→ Tagged
→ In Transit (technician's complete_tagging confirmation moves it here directly)
→ Arrived (physical unlock) | Awaiting Seal Return (remote unlock)
→ Untagging In Progress → Untagged → Awaiting Seal Return (physical path only)
→ Completed
→ Cancelled

Journey Request (`journey_request_status`):

Draft → Pending Control Room Approval → Tagging
→ Journey Ready → Untagging → Awaiting Seal Return → Seal Returned
(Rejected / Cancelled at approval gates)

Tagging Booking (`booking_status`):

Draft → Pending Account Manager Review → Pending Finance PCB Approval
→ Finance PCB Approved / Finance PCB Rejected → Cancelled

PCB Job Order (`job_order_status`):

Unassigned → Team Leader Assigned → Completed / Cancelled

PCB Assignment (`assignment_status`):

Pending → Assigned
→ Pending Untagging Assignment → Untagging Assigned
→ Pending Seal Return Assignment → Seal Return Assigned
→ Cancelled

Seal Device (`current_status`):

Quality Check → Available → Assigned → In Journey → Arrived → Untagged → Returned
(or Damaged / Lost / Inactive at any point)

Cancellation / Rework Paths

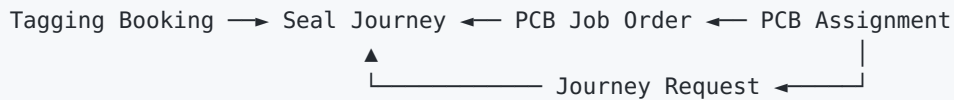
Cancellation and rejection are possible at nearly every stage, guarded by checks that block reversal once real field work has started:

- `reject_booking` / `reopen_booking` — before Finance approval.
- `reject_by_control_room` — before tagging begins.
- `_cancel_assignment` on PCB Assignment — unwinds the whole chain: Journey Request → `Cancelled`, Tagging Booking → back to `Pending Finance PCB Approval`, Seal Journey → back to `Pending Finance PCB Approval`. Only allowed before tagging has actually started (`_TAGGING_STARTED_JOURNEY_STATUSES` guard).

Audit Trail

Every action on a Journey Request (submitted, approved/rejected, tagging completed, untagging confirmed, seal return confirmed, etc.) is logged with actor, timestamp, and remarks in the **Journey Request Approval** child table — this is the human-readable history of the journey.

Doctype Relationships



- `Tagging Booking.seal_journey_reference` → Seal Journey
- `Tagging Booking.pcb_job_order_reference` → PCB Job Order
- `PCB Job Order.tagging_booking` → Tagging Booking
- `PCB Job Order.assignment_reference` → PCB Assignment
- `PCB Assignment.pcb_job_order / seal_journey / tagging_booking` → respective docs
- `Journey Request.job_order` → PCB Job Order; `journey_reference` → Seal Journey
- `Seal Journey.tagging_booking / pcb_job_order / pcb_assignment / journey_request` → all four upstream docs (mirror source of truth)